Unit-IV

4. (a) Write the application of SIM instruction. 2

(b) What do you mean by vector interrupt? Explain 8085 vector & interrupts with diagram. 7

(c) How to implement RST 5? Explain with diagram. 7

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(d) Design a 1 minute time using a 60 Hz power line is an interupting source. The output ports should display minutes and second in BCD. At the end of the minute, the output ports should continue displaying one minute and zero seconds. 7

Unit-V

5. (a) Write the CWR format of 8155 PPI. 2

(b) Draw block diagram of USART (8251) and explain. 7

(c) What is BSR control word format of 8255? Write a BSR control word subroutine to set bit PC_7 and PC_3 and reset them after 10 ms. Assume that delay subroutine is available. 7

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(d) Write a program to generate a 1 kHz square wave from counter 1 of 8253. Assume the gate is tied with + 5 volt (counter 1). Explain the significance of connecting the gate to + 5 volt. 7

328515(28)

B. E. (Fifth Semester) Examination, April-May 2016

(Old Scheme)

(AEI, CSE, EEE, EI, Et & T, IT & Mechatronics Engg. Branch)

MICROPROCESSOR & INTERFACES

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Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : Part (a) of each question is compulsory. Solve any two questions from (b), (c) and (d) of each unit. All questions are compulsory.

Unit-I

1. (a) Why not possible more than 256 instructions in 8085 microprocessor? 2

(b) How to demultiplex the AD_0 to AD_7 lines of 8085 microprocessor? Write the advantages &

disadvantages of multiplexing of Address/Data Bus (AD_0 to AD_7) of microprocessor. 7

(c) Explain the function of following pins of 8085 microprocessor : 7

(i) TRAP

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(ii) HOLD

(iii) IO/$\overline{M}$

(iv) SOD

(v) READY

(d) Draw the timing diagram of Instruction MVI M, data. 7

Unit-II

2. (a) What is PSW? 2

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(b) A set of ten current reading is stored in memory location starting at XX70H. The reading are expected to positive ($<127_{10}$). Write a program to :

(i) Check each reading to determine whether positive or negative.

(ii) Reject all negative readings.

(iii) Add all positive readings.

(iv) Output FFH at port 1 at any time when the sum exceeds eight bits to indicate overload; otherwise display the sum. If no output port is available in

the system go to step (v).

(v) Store FFH in the memory location XX70 H when the sum exceed eight bits otherwise store the sum.

Data (*H*) 28, D8, C2, 21, 24, 30, 2F, 19, F2, 9F. 7

(c) Binary number is stored in memory location 4260 H. Write a program to convert binary no. into ASCII Hex code and store in memory locations 4150 H and 4151 H. 7

(d) Explain the addressing model of 8085 microprocessor with examples. 7

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Unit-III

3. (a) How to generate $\overline{IORD}$ & $\overline{IOWR}$ control signals? 2

(b) Write the difference between memory mapped I/O and I/O mapped I/O. 7

(c) Explain Microprocessor controlled & Peripheral controlled data transfer. 7

(d) Interface following memory chips with 8085 microprocessor : 7

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(i) (4 K × 8) EPROM 1 No. starting address 0000H

(ii) (8 K × 8) RAM 1 No. starting address 1000H

(iii) (4 K × 4) SRAM 2 No. starting address 4000H

PTO